

Physics Waec Questions Candidate/Private

Question 1

- (a) Briefly explain the following terms:
- (i) emission line spectra;
 - (ii) line absorption spectra.
- (b) Draw a labelled diagram showing the structure of a simple type of photocell and explain its mode of operation.
- (c) State two
- (i) reasons to show that x-rays are waves
 - (ii) uses of x-rays other than in medicine.
- (d) An electron jumps from an energy level of -1.6eV to one of -1.4 eV in an atom. Calculate the energy and wavelength of the emitted radiation. [$h = 6.6 \times 10^{-34}\text{ Js}$; $c = 3.00 \times 10^8\text{ ms}^{-1}$; $\text{eV} = 1.6 \times 10^{-19}\text{ J}$]

Question 2

- (a) State two essential differences between a moving coil galvanometer and a d.c. generator.
- (b) Explain the term eddy currents and state two devices in which the currents are applied.
- (c) State the principle on which the potentiometer is based when it is functioning.
- (d) A source of e.m.f. 110 V and frequency 60Hz is connected to a resistor, an inductor and a capacitor in series. When the current in the capacitor is 2A , the potential difference across the resistor is 80 V and that across the inductor is 40 V . Draw the vector diagram of the potential differences across the inductor, the capacitor and the resistor.

Calculate the:

- (i) potential difference across the capacitor.
- (ii) capacitance of the capacitor.
- (iii) inductance of the inductor. [$\pi = 3.14$]

Question 3

1. A projectile is fired at an angle to the horizontal with velocity, u . Show that at any time, t , during the motion, the:
 - The horizontal component of the velocity is independent of t ;
 - The vertical component of the velocity depends on t .
2. State the assumption on which projectile motion is based.

Question 4

State three differences between geostationary satellites and polar satellites.

Question 5

1. Using the kinetic theory, explain the term.
2. Name one phenomenon that demonstrates that light behaves as a:
1.Wave; ii.Particle.

Question 6

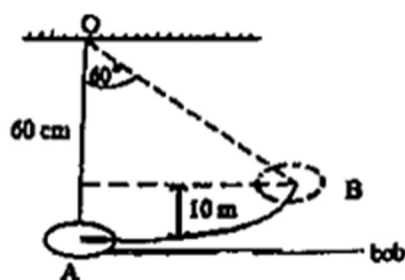
1. State one difference between an intrinsic and an extrinsic semiconductor.
2. Draw a circuit diagram to illustrate full-wave smoothing rectification.

Question 7

- (a) What is fibre optics?
- (b) State two reasons why optical fibres are preferred to copper cables in the telecommunication industry.

Question 8

- (a) (i) State the reason why simple harmonic motion is periodic.
- (ii) State two factors that affect the period of oscillation of a simple pendulum.
- (iii) Sketch a graph of the total mechanical energy, E , against displacement, y , for the motion of a simple pendulum from one extreme position to the other.
- (b) The diagram below illustrates an oscillatory pendulum.



Calculate the work done in raising the pendulum to point B, if the mass of the bob is 50g. [$g = 10 \text{ ms}^{-2}$]

1. A spiral spring of spring constant, k , and natural length, l , has a scale pan of mass 0.04 kg hanging on its lower end while the upper end is firmly fixed to a support. When an object of mass 0.20 kg is placed on the scale pan, the length of the spring becomes 0.055 m and when the object is replaced with another object of mass 0.28 kg, the length of the spring becomes 0.065 m. Calculate the values of k and l .

[$g = 10 \text{ ms}^{-2}$]

Question 9

(a) Define each of the following terms used with simple machines:

(i) Pivot;

(ii) Load;

(iii) Efficiency.

(b) A truck of mass 1.23 kg is pulled from rest by a constant horizontal force of 25.2 N on a leveled road. If the maximum speed attainable in the process is 60 km/h:

(i) Work done by the force;

(ii) Distance travelled by the truck in reaching the maximum speed.

(c) State two differences between absolute zero temperature and ice point.

(d) An uncalibrated liquid-in-glass thermometer was used in determining a Celsius temperature. The readings are tabulated below.

Temperature/ $^{\circ}\text{C}$	-6	0	100
Length of column/cm	L	2.0	15.0

Determine the value of L .

Question 10

(a) (i) Why are parabolic mirrors suitable for use in headlamps of vehicles?

(ii) Draw a ray diagram to illustrate the answer in 10(a)(i).

(b) (i) State two applications of echoes.

(ii) An observer standing at a point, P, on the same horizontal ground as the foot, H, of a tower shouts and 1.20 s later, he hears the echo. He then moved to another point, Q, 40 m from P, and shouts again but the echo was heard after 1.45 s. Calculate the:

(i) Distance between P and H;

(ii) Speed of sound in air.

(c) (i) Define the term absolute refractive index of a medium.

(ii) A piece of coin falls accidentally into a tank containing two immiscible liquids A and B as illustrated in Fig. 10.0.

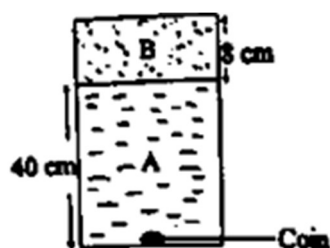


Fig. 10.0

Calculate the displacement of the coin when viewed vertically from above.
[refractive index of A = 1.3, refractive index of B = 1.4]

Question 11

- State the function of each of the following parts of a modern X-ray tube:
 - Heater;
 - High tension source;
 - Cooling fins.
- State one reason for each of the following design features of a modern X-ray tube:
 - The glass envelope is highly evacuated;
 - The target is a metal with a very high melting point;
 - The cooling fins are located outside the glass envelope.
- In a nuclear fission reaction, a nuclide ${}_{92}^{235}\text{U}$ is bombarded with a neutron to

produce ${}_{56}^{140}\text{Ba}$ and ${}_{36}^{94}\text{Kr}$ with additional neutrons. The energy involved in the process is Q.

- Write down the balanced nuclear reaction equation for the process.
- State with reason whether Q is absorbed or released in the process.
- Calculate the value of Q in joules.
- State three differences between nuclear fusion and nuclear fission.

Question 12

A particle is projected at an angle of 60° to the horizontal with a speed of 25 m/s. [$g = 10 \text{ m/s}^2$].

Calculate the:

- Time of flight of the particle;
- Horizontal component of the speed of the particle at its maximum height.

Question 13

List three applications of thermionic emission.

Question 14

Explain briefly why the Young's modulus of aluminum is greater than that of rubber.

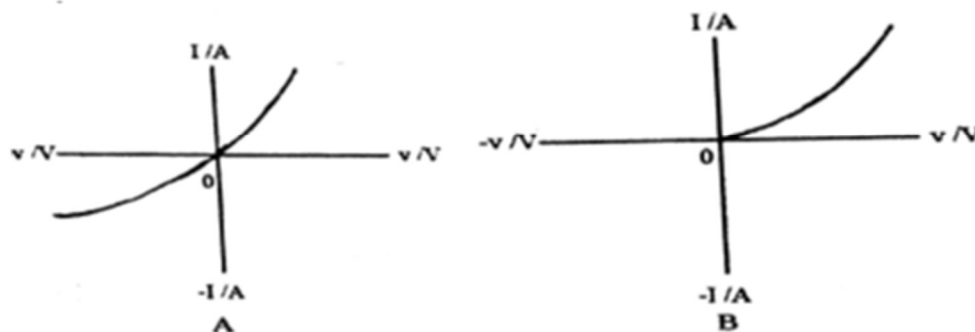
Question 15

The expression for the period of oscillation, T , of a mass-spring system in terms of the mass, m , on the spring and force constant, k , is given as $T = b \sqrt{\frac{k}{m}}$.

Where b is a dimensionless constant. Prove the validity or otherwise of the expression using dimensional analysis.

Question 16

The output characteristics of two diodes A and B are illustrated in the diagrams below.



- (a) Which of the diodes is an ideal rectifier?
- (b) State the reason for the answer given in 5(a).

Question 17

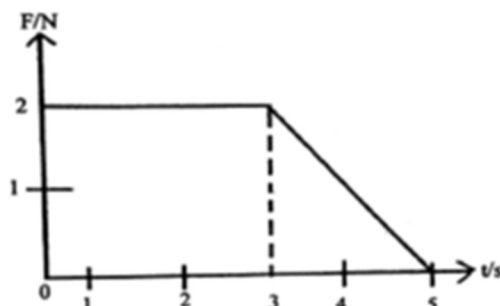
- (a) State two factors on which the pressure at a point in a fluid depends.
- (b) A uniform cylindrical vessel has a cross-sectional area of 120 cm^2 . It is filled to a depth of 150 cm with water and oil of thickness 50 cm floats on the water. [$g = 10 \text{ m/s}^2$, density of oil = 700 kg/m^3 , density of water = 1000 kg/m^3 , Atmospheric pressure = $1.013 \times 10^5 \text{ N/m}^2$]

Calculate the:

- (i) Absolute pressure at the base of the cylinder;
- (ii) Force acting on the base of the cylinder.

- (c) (i) State one difference between a resultant force and an equilibrium force.
(ii) Write down the equation for the maximum acceleration of a body oscillating with amplitude r . Identify the terms used in the equation.

(d)



The diagram above shows a force-time graph for an object of mass 1.5 kg. Determine the:

- (i) Impulse of the force;
(ii) Final velocity of the object if it was initially at rest.

Question 18

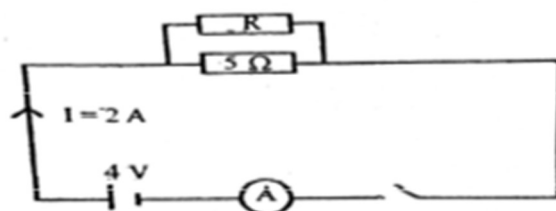
- (a) Give three examples of thermometric properties.
(b) A quantity of water of mass 20 g at -5°C was gradually heated until it finally boils.
(i) Sketch a graph of density against temperature for the heating process.
(ii) Calculate the minimum energy required to melt the ice. [Specific heat capacity of ice = $2.1 \times 10^3 \text{ J/kg}$, Specific latent heat of fusion of ice = $3.2 \times 10^5 \text{ J/kg}$]
(c) Define each of the following terms associated with a simple machine:
(i) Load;
(ii) Velocity ratio.
(d) State whether or not velocity ratio is affected by frictional forces and give the reason for your answer.

Question 19

A missile is projected so as to attain its maximum range. Calculate the maximum height attained if the initial velocity of projection is 200 m/s. [$g = 10 \text{ m/s}^2$]

Question 20

- (a) (i) Define the term neutral points as used in magnetism.
- (a) (ii) Draw a diagram showing the pattern of a magnetic field around a bar magnet.
- (b) A particle of charge 3.2×10^{-19} C enters a uniform magnetic field with a speed of 5.0×10^7 m/s. If the magnetic field strength is 8 T calculate the maximum force that can be experienced by the particle.
- (c) (i) State the condition necessary for resonance in an R-L-C series circuit.
- (c) (ii) State one application of resonance in R-L-C circuits.
- (d)



- (d) The diagram above illustrates an electrical circuit diagram. Use it to calculate the:
- (i) Value of R
- (ii) Current through R.

Question 21

- (a) State two applications of radioactivity in the field of agriculture.
1. Pest Control: Radioactive isotopes are used to sterilize insects, which helps in controlling pest populations by preventing them from reproducing.
 2. Soil and Water Management: Radioactive tracers are used to study soil movement and water distribution patterns, which helps in efficient irrigation and soil conservation practices.
- (b) The table below provides information on the decay of a radioactive material.

Time (min)	0	20	40	60	80	100	120	140
Count rate (min ⁻¹)	21000	9300	4300	2000	1300	1000	830	660

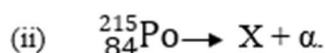
(i) **Plot a graph of count rate on the vertical axis and time on the horizontal axis.**

- Use a graph sheet and plot the given data points.
- Draw a smooth curve through the points to represent the decay process.

(ii) Use the graph to determine the half-life, $T_{\frac{1}{2}}$ of the radioactive material.

(iii) Calculate the decay constant.

(c) Identify the nuclide, X, in the following decay processes:



(ii) **Calculate the half-life of the radioactive material.**

1. Identify two points on the graph where the count rate halves.
2. Measure the time interval between these points to find the half-life.

Question 22

- (a) What does the acronym *LASER* stand for?
(b) State **two** areas of application of *LASERS*.

Question 23

- (a) (i) What are *crystalline substances*?
(ii) State **two** examples **each** of:
(α) *crystalline substance*
(β) *amorphous substance*
- (b) (i) A body was placed at different points on the earth's surface and it was observed that its weight varies at different points. State **three** reasons for the observation.
(ii) A car travels from a starting point, P, and moves x km northwards, 40 km eastwards, 30 km southwards and then gets to a point M where it turns 40° west of south at a distance of 50 km from its starting point, P.
(α) Sketch a diagram showing the directions of motion of the car at **each** stage.
(β) Calculate the value of x .
- (c) State **three** characteristics of pressure in fluids

Question 24

- (a) (i) The *s.v.p.* of water vapour on a certain day at air temperature 28°C is 12.2 mmHg. Calculate the relative humidity of a day when the *s.v.p.* of water vapour at dew point is 15.9 mmHg.
- (ii) State **two** effects of expansion of water when it freezes to ice.
- (b) A quantity of liquid of mass 600 g placed in a plastic container with a tight fitting lid has a temperature of 30°C . The container is placed in a microwave oven rated 1250 W.
- (i) If the microwave is operated for 4 minutes, calculate the **final** temperature attained by the liquid. (*assuming no heat losses*) [Specific heat capacity of the liquid = $4100 \text{ J kg}^{-1} \text{ K}^{-1}$]
- (ii) If the liquid is brought out and allowed to cool, a dent is observed on the container. Explain.
- (c) Explain why containers with tight-fitting lids are **not** suitable for use in microwave cooking.
- (d) (i) Define *fixed points* on a temperature scale.
- (ii) State **one** effect of heat on a substance.

Question 25

- (a) State **two** conditions necessary for diffraction of waves to occur.
- (b) Explain **each** of the following terms as used in connection with waves:
- wavefront*;
 - wavelength*;
 - antinode*.
- (c) Two plane progressive waves, $y_1 = 0.2 \sin(200\pi t)$ and $y_2 = 0.3 \sin(200\pi t)$, where y and x are both in cm and t in s, undergo superposition. Sketch the
- displacement-time graph of y_1 for **one** complete cycle;
 - resultant wave, Y , for the superposition of y_1 and y_2 .
- (d) Draw a ray diagram illustrating the operation of a simple microscope.

Question 26

- (a) State **two** uses of electromagnets.
- (b) A resistance wire has a length of 1.5 m and cross-sectional area of $7.85 \times 10^{-7} \text{ m}^2$. If the wire is connected across the left gap of a meter bridge and a 5

Ω resistor across the right gap, a balance point of 60.0 cm from the left side of the bridge wire is obtained. Calculate the resistivity of the wire.

(c) The plates of a parallel plate capacitor, each of area 2.0 m^2 , are $5.0 \times 10^{-3} \text{ m}$ apart in air. If the p.d. across the plates is 10 kV, calculate the:

- (i) capacitance of the capacitor;
- (ii) charge on **each** plate;
- (iii) electric field intensity across the plates.

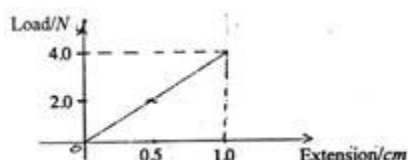
$$[\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}]$$

(d) (i) Define *electrical potential* at a point in an electric field.

(ii) State the relationship between *electric potential* (V) and *electric field intensity*

Question 27

The load-extension graph of an elastic material is illustrated below.



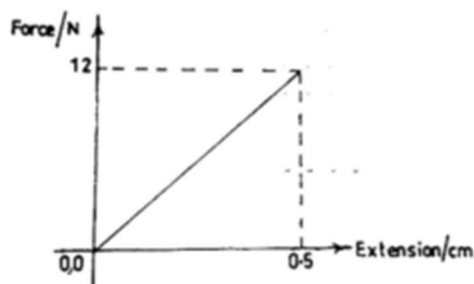
Use the graph to determine the work done in stretching the material.

Question 28

- (a) State the scientific principle underlying the operation of fibre optics.
- (b) Explain each of the following terms as used in fibre optics: (i) core; (ii) cladding.

Question 29

- (a) Define uniform acceleration.
- (b) Forces act on a car in motion. List the: (i) horizontal forces and their directions; (ii) vertical forces and their directions.
- (c) A car starts from rest and accelerates uniformly for 20s to attain a speed of 25 m/s. It maintains this speed for 30s before decelerating uniformly to rest. The total time for the journey is 60s. (i) Sketch a velocity-time graph for the motion. (ii) Use the graph to determine the: (a) total distance travelled by the car; (b) deceleration of the car.



(d)

Question 30

- (a) List **two** factors **each** that affect heat loss by: (i) radiation; (ii) convection.
- (b) State **two** factors that determine the quantity of heat in a body.
- (c) Explain the statement: The specific latent heat of vaporization of mercury is $2.72 \times 10^5 \text{ J kg}^{-1}$.
- (d) A jug of heat capacity 250 J/K contains water at 28°C . An electric heater of resistance 35Ω connected to a 220 V source is used to raise the temperature of the water until it boils at 100°C in 4 minutes. After another 5 minutes, 300 g of water has evaporated. Assuming no heat is lost to the surroundings, calculate the: (i) mass of water in the jug before heating; (ii) specific latent heat of vaporization of steam.

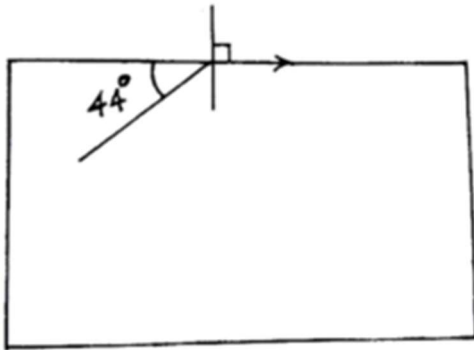
[Specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$]

Question 31

- (a) Define diffraction.
- (b) (i) Explain critical angle.

The diagram below illustrates a ray of light passing through a rectangular transparent plastic block.

(ii)



(α) Determine the value of the critical angle.

(β) Calculate the refractive index of the block.

(c) A pipe closed at one end has a fundamental frequency of 200 Hz. The frequency of the first overtone of the closed pipe is equal to the frequency of the first overtone of an open pipe. Calculate the: (i) fundamental frequency of the open pipe; (ii) length of the closed pipe; (iii) length of the open pipe.

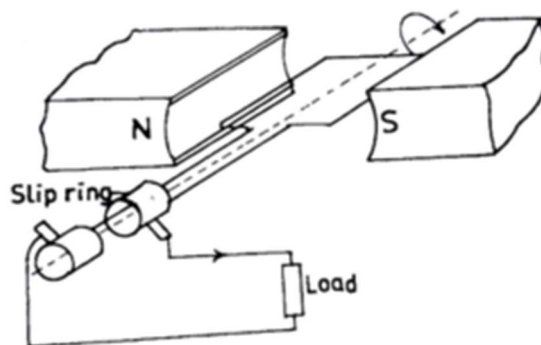
[Speed of sound in air = 330 m/s]

Question 32

(a) Define: (i) reactance; (ii) impedance in an a.c. circuit.

The diagram below illustrates an a.c. generator. When the coil is rotated, an e.m.f. is induced in the coil.

(b)



(i) Explain why an e.m.f. is induced.

(ii) State the purpose of the slip-rings.

(iii) Name and state the law used to determine the direction of the induced current.

(iv) State **two** ways to increase the induced e.m.f.

(c) A lamp is rated 12V, 6W. Calculate the amount of energy transformed by the lamp in 5 minutes.

Question 33

(a) Define binding energy in an atom.

(b) List three evidences to support the claim that X-rays are electromagnetic waves.

(c) List three peaceful uses of nuclear energy.

(d) Light of wavelength $4.5 \times 10^{-7} \text{ m}$ is incident on a metal resulting in the emission of photoelectrons. If the work function of the metal is $3.0 \times 10^{-19} \text{ J}$, calculate the:

(i) frequency of the incident light;

(ii) energy of the incident light;

(iii) energy of the photoelectrons.

[Speed of light = $3.0 \times 10^8 \text{ m/s}$, $h = 6.6 \times 10^{-34} \text{ Js}$]

Question 34

A tennis ball travelling with a speed of 10m/s rolls off a horizontal table 1.0m high. Ignoring air resistance, determine the:

1. horizontal speed of the ball just as it strikes the floor;
2. time of flight of the ball in air.

[$g = 10 \text{ m/s}^2$]

Question 35

A ball is projected vertically upwards with a velocity of 20m/s.

1. Calculate the height attained 4 s after projection.
2. What inference can you draw from your answer?

[$g = 10 \text{ m/s}^2$]

Question 36

- (a) (i) What is nuclear fission?
 (ii) State the function of **each** of the following materials in a nuclear fission reactor:
 (α) graphite;
 (β) boron rod;
 (γ) liquid sodium

- (b) The table below gives some of the energy levels of a hydrogen atom.

N	1	2	3	4	5	∞
E_n/E_V	-13.60	-3.39	-1.51	-0.85	-0.54	0.00

- (i) Draw the energy level diagram for the atom.
 (ii) Determine the wavelength of the photon emitted when the atom goes from the energy state $n = 3$ to the ground state.
 [$h = 6.6 \times 10^{-34} \text{ Js}$, $c = 3.0 \times 10^8 \text{ ms}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$]
- (c) A piece of ancient bone from an excavation site showed $^{14}_6\text{C}$ activity of 9.5 disintegrations per minute per $1.0 \times 10^{-3} \text{ kg}$. If a bone specimen from a living creature shows $^{14}_6\text{C}$ activity of 12.0 disintegrations per minute per $1.0 \times 10^{-3} \text{ kg}$, determine the age of the ancient bone. [Half-life of $^{14}_6\text{C} = 5572 \text{ years}$]

Question 37

1. (i) What is a dielectric?
 (ii) A parallel plate capacitor consists of two plates each of area $9.6 \times 10^{-3} \text{ m}^2$, separated by a dielectric of thickness $2.25 \times 10^{-4} \text{ m}$ and dielectric constant 900. Calculate the capacitance of the capacitor.
 [$\epsilon_0 = \text{permittivity of free space} = 8.85 \times 10^{-12} \text{ F/m}$].
2. (i) Which of the following devices has a higher resistance: an ammeter or a voltmeter? Give a reason for your answer.
 (ii) The resistance of the voltmeter in the circuit diagram illustrated below is 800Ω . Calculate the voltmeter reading.

Question 38

(a) Define each of the following terms as applied to the structure of the atom:

(i) isotopes;

(ii) isobars.

(b) A lead sphere of mass $5.0 \times 10^{-3} \text{ kg}$ is allowed to fall to the ground from a height of 20.0 m.

Determine the:

(i) speed of the sphere just before it hits the ground;

(ii) de Broglie wavelength of the sphere just before it hits the ground. [$g = 10.0 \text{ m/s}^2$; $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$]

(c) An electron is accelerated in an X-ray tube with a supply of 100 kV accelerating potential.

Calculate the:

(i) speed of the electron;

(ii) wavelength of the X-ray that will be produced when the electron is suddenly stopped by a solid metal placed in its path. [$h = 6.63 \times 10^{-34} \text{ J} \cdot \text{s}$; $e = 1.6 \times 10^{-19} \text{ C}$; $m_e = 9.1 \times 10^{-31} \text{ kg}$]

Question 38

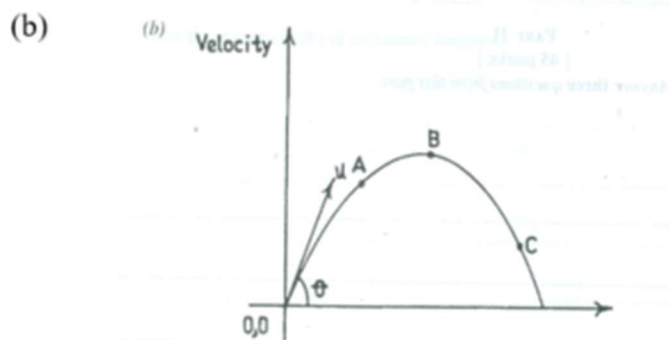
A tennis ball travelling with a speed of 10 m/s rolls off a horizontal table 1.0m high. Ignoring air resistance, calculate the magnitude of the vertical component of the velocity of the ball as it strikes the floor. [$g=10\text{m/s}^2$]

Question 39

An electron with speed of $1.8 \times 10^5 \text{ ms}^{-1}$ moves in a circular path, the plane of which is perpendicular to a uniform magnetic field of $1.0 \times 10^{-5} \text{ T}$. Calculate the radius of the orbit of the electron. [Mass of electron = $9.1 \times 10^{-31} \text{ kg}$, charge on electron = $1.6 \times 10^{-19} \text{ C}$]

Question 40

(a) Define: (i) terminal velocity (ii) linear momentum



(b) The diagram above illustrates the trajectory of a projected object. Neglecting friction, how does the:

1. horizontal component U_x of the initial velocity u change at the points **A**, **B**, **C**?
2. vertical component U_y of the initial velocity u change at the points **A**, **B**, **C**?

(c) A uniform bar **XY** of mass 5.0kg and length 300cm is suspended by two inextensible identical wires **P** and **Q**. **P** is attached 50cm from the end **X** and **Q** to the centre of mass **C** of the bar, such that they are vertical. A mass of 10.0kg is attached to the end **X** of the bar.

1. Draw and label a diagram of the arrangement.
2. Determine the distance from the centre of mass **C** where another mass of 10.0kg should be hung on the bar to keep the bar horizontal and to make the tension in **Q** four times the tension in **P**. [$g=10 \text{ m/s}^2$]

Question 41

- (i) State Boyle's law.
 - (ii) State **two** assumptions of the kinetic theory of gas necessary for a real gas to be considered as ideal.
- (i) Explain the term renewable energy.
 - (ii) List **four** examples of renewable energy.
- ☐ Why is the maximum arm of the Six's maximum and minimum thermometer **not** completely filled with alcohol whereas the minimum arm is filled?
 - ☐ An electric heater rated 100W is used to raise the temperature of 200g of water in a flask from 30°C to 100°C. The water is boiled for a further 2 minutes before the heater is switched off. Neglecting heat losses, calculate the:
 - (i) Quantity of energy supplied to the water in raising its temperature to 100°C.
 - (ii) Mass of steam that escaped. [Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$; specific latent heat of vaporization of steam = $3.3 \times 10^6 \text{ J kg}^{-1}$]

Question 42

1. (i) State **three** differences between sound and light waves.
2. (i) Sketch and label a ray diagram to illustrate the image formed by a converging lens if an object is situated at twice the focal length of the lens.
 - (ii) State the nature, position, and size of the image formed in (b)(i).
3. State why an observer **cannot** see around corners.

4. In a game reserve, the converging lens of the camera used to observe a giraffe 36 m away forms an inverted image of height 2.5 cm at 0.030 cm behind the lens.

Calculate the: (i) magnification produced; (ii) height of the giraffe; (iii) focal length of the lens; (iv) power of the lens.

Question 43

- State **two** factors that determine the sensitivity of a moving-coil galvanometer.
- Define the farad.
- (i) What happens in a series circuit if more bulbs are added in series?
(ii) A laboratory technician wishes to reduce the current passing through a galvanometer to 0.05 of its initial value by means of a resistor. If the internal resistance of the galvanometer is 100Ω :
 - (α) how should the resistor be connected?
 - (β) calculate the resistance of the resistor.
- The current I in a sinusoidal a.c. circuit is given by $I = 40\sin(500\pi t)$ where t is the time in seconds. Use the equation to determine the:
 - frequency;
 - peak value;
 - r.m.s. value of the current.

Question 44

- (i) Define decay constant.
(ii) An isotope of Radon has a half-life of 3.8 days. Calculate its decay constant.
- Define the following terms: (i) Mass defect; (ii) Binding energy of a nucleus.
- State the function and material used for **each** of the under listed components in a nuclear fission reactor: (i) Moderator; (ii) Control rods; (iii) Coolant.
- A nucleon of hydrogen-3 is formed when a neutron is absorbed by a nucleus of hydrogen-2. The table below gives the masses of the particles in atomic mass unit (u):

Particle	${}_0^1n$	${}_1^2H$	${}_1^3H$
Mass/u	1.00867	2.00141	3.00160

- (i) Write a balanced equation for the reaction.

6. (ii) Calculate the energy released in this reaction.

Question 45

State the dimensions of the following physical quantities:

1. Acceleration
2. Pressure
3. Density

Question 46

A projectile is launched with a velocity of 42m/s at an angle of 30° to the horizontal. If the time of flight is 4.2 s, calculate the maximum attainable height. (10m/s^2)

Explain why mercury does **not** wet glass while water does.

Question 47

The table below shows the pointer reading of a spring balance when different weights are hung on it. Determine the value of FFF.

Weight/N	5	10	FFF
Pointer reading/cm	8	12	30

Question 48

Explain the following terms:

1. Tensile stress
2. Young's modulus

Question 49

1. (i) Explain **relative motion**.
(ii) Two cars moving in opposite directions on the same straight road with velocities 80 km/h and 60km/h respectively pass each other at a point. Determine the velocity of the first car relative to the second car.
2. (i) Define **force**.
(ii) Classify the following forces as either **contact forces** or **field forces**: push; tension; gravitational force; electrostatic force; reaction; and magnetic force.
3. A body starts from rest and travels in a straight line for 2s with uniform acceleration of 1m/s^2 . It then travels at a constant speed for some time before coming to rest with a uniform retardation of 2m/s^2 . The total distance covered by it is 15m

(i) Draw and label a velocity-time graph for the motion.

(ii) Calculate the:

- (α) velocity attained at the end of 2s;
- (β) total time taken for the journey.

Question 50

1. Name the energy transformations which occur in a functioning:

- Car engine
- Steam turbine

2. List the **three** main features of a clinical thermometer.

3. Explain why water becomes cooler when kept in a clay pot for some time in a room.

4. Water of mass 0.4 kg loses 210 kJ of energy when taken from the Earth's surface to Mars with a surface temperature of -60°C . The water temperature decreases until the water freezes. The temperature of the ice formed decreases until it becomes equal to that on Mars. Calculate the temperature of the water while on Earth.

[Specific latent heat of fusion of ice = $3.3 \times 10^5 \text{ J/kg}$; specific heat capacity of water = 4200 J/kg K^{-1} ; specific heat capacity of ice = 2100 J/kg K^{-1}]

Question 51

State **two** differences between a **sound wave** and a **radio wave**.

Explain why a vibrating tuning fork sounds louder when its stem is pressed against a table top than when held in air.

State **two** conditions necessary for the:

- Production of a stationary wave in a medium.
- Formation of interference wave patterns.
- Occurrence of total internal reflection of a wave.

A ray of light is incident on **one** face of an equilateral glass prism.

- Draw a ray diagram to show the path of the ray through the prism.
- Calculate the refractive index of the glass if the angle of minimum deviation is 41° .

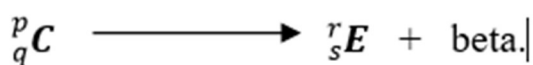
Question 53

Part II. Candidates are expected to answer any **three Questions** from this part.

1. State the terms:

- **Transmutation** as it relates to radioactivity;

- **Stopping potential.**



A nucleus **C**, formed artificially from **A** and **B**, is radioactive and quickly decays to another nucleus **E** as indicated in the nuclear equations below.

Determine the values of **p**, **q**, **r**, and **s**.

A certain metal of work function 1.6 eV is irradiated with an ultraviolet light of wavelength 3.6×10^{-7} m. Calculate the maximum:

- Kinetic energy of an ejected electron in joules;
- Speed of an emitted electron.

(Given: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$, $c = 3.0 \times 10^8 \text{ ms}^{-1}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$, $h = 6.6 \times 10^{-34} \text{ Js}$)

If the source of the ultraviolet light in 12(c) is moved away from the surface of the metal, state its effect on the maximum speed of the ejected electron.

Question 53

- Define electromotive force.
- State:
the principle of operation of a potentiometer;
two advantages that a potentiometer has over a voltmeter in measuring potential difference.
- Sketch and label a diagram of a gold-leaf electroscope
 - Give one use of a gold-leaf electroscope
- Explain the action of a magnetic relay:
 - List two factors which determine the magnitude of an induced emf in a coil.
 - A current of 5 A passes through a straight wire in a uniform magnetic field of flux density $2.0 \times 10^{-3} \text{ T}$. Calculate the force per unit length exerted on the wire when it is inclined at 30° to the field.

Question 54

- State the three characteristics of sound and the factor on which each of them depends.

- (b) Explain resonance as applied to sound.
- (c) What role does echo play in the construction of a concert hall?
- (d) The surface of an ear drum (assumed circular) has a radius 2.1 mm. It resonates with an amplitude of 0.8×10^{-7} m as a result of impulses received from an external body vibrating at 2400 Hz. If the resulting pressure change on the ear drum is 3.6×10^{-5} Nm⁻² calculate the:
 - (i) period of oscillation.
 - (ii) velocity:
 - (iii) acceleration:

force.

[$\pi = 3.14$]

Question 55

- (a) Define heat capacity and state its unit.
- (b) List two effects of heat on a substance
- (c) Explain how a tightly fitted glass stopper could be removed from a reagent bottle.
- (d) A quantity of pepper soup of mass 800g poured into a plastic container with a tight fitting lid has a temperature of 30°C
The container is then placed in a microwave oven, rated 1200 W and operated for 3 minutes.

Calculate the final temperature attained by the soup. (Assuming no heat losses)

Explain why containers with tight-fitting lids are not suitable for use in microwave cooking.

When the soup is brought out and allowed to cool, a dent was observed on the container. Explain.

[Take specific heat capacity of the soup = $4000 \text{ J kg}^{-1} \text{ K}^{-1}$]

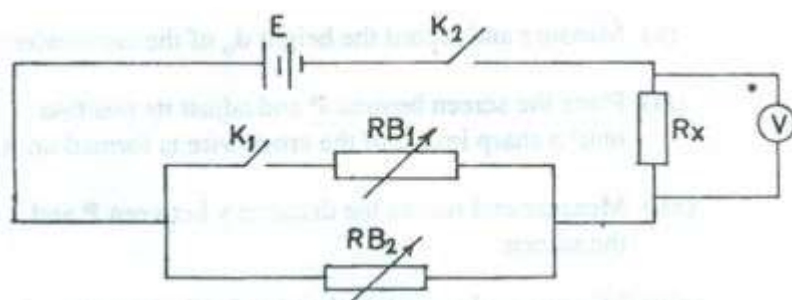
Question 56

- (a) Give two examples each of:
 - (i) rotational motion;
 - (ii) linear motion

(b) Describe a laboratory experiment to determine the density of an irregularly shaped solid.

(c) State Newton's second law of motion.

(d) Explain the term inertia.



(e)

The diagram above illustrates a body of mass 5.0 kg being pulled by a horizontal force F . If the body accelerates at 2.0 ms^{-2} and experiences a frictional force of 5 N , calculate the:

(i) net force on it;

(ii) magnitude of F ;

(iii) coefficient of kinetic friction

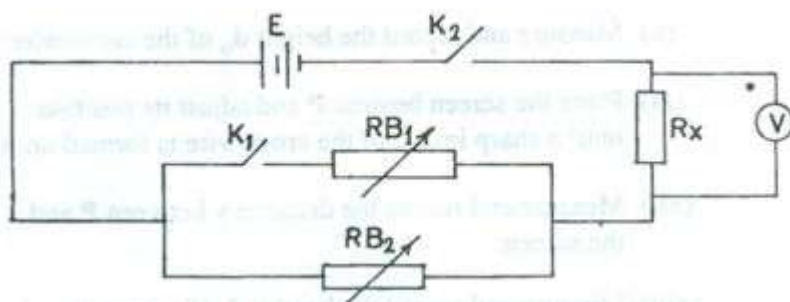
$$[g = 10 \text{ ms}^{-2}]$$

Question 57

(a) State two conditions under which photo-electrons can be emitted from the surface of a metal.

(b) List two particles characteristics of electromagnetic waves.

Question 58



The diagram above represents the graph of electron energy against the frequency of the radiation incident on a metal surface.

Interpret the:

- slope of the graph;
- intercept, OC;
- intercept, OK.

Question 59

A metallic bar 50 cm long has a uniform cross-sectional area of 4.0 cm^2 . If a tensile force of 35 kN produces an extension of 0.25 mm, calculate the value of Young's modulus.

Question 60

Explain why it is possible for a slow moving neutron to penetrate a nucleus while an alpha particle needs a larger amount of energy to penetrate the same nucleus.

List four properties of cathode rays.

In a photoelectric emission, a faint blue light of wavelength $5.0 \times 10^{-7} \text{ m}$ and a bright red light of wavelength $7.0 \times 10^{-7} \text{ m}$ are incident on a metal plate. Which of the radiations will release:

greater number of electrons;

electrons of higher maximum kinetic energy? Give reasons for your answers

Caesium metal of work function 2.0 eV is illuminated with radiations of wavelength $5.0 \times 10^{-7} \text{ m}$. Calculate the:

threshold wavelength of the metal;

maximum energy of the liberated electrons.

[$h = 6.6 \times 10^{-34} \text{ Js}$; $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$; $c = 3.0 \times 10^8 \text{ ms}^{-1}$]

Question 61

- What is a 'lightning conductor'?
- Why are lightning conductors used to protect tall buildings from the undesirable effect of thunderstorms?

Draw the lines of force associated with the following situations:

Two unlike point charges of equal magnitude;

A positive charge in isolation;

Two parallel plates carrying unlike charges of equal magnitude.

(c) Two equal point charges exert a force F on each other when they are $5.0 \times 10^{-2} \text{ m}$

apart in a vacuum. Determine the distance between the charges if the force is to be doubled.

Question 62

- (a) List three effects of heat on matter
- (b) Explain the following observations:

a thick glass cup cracks when boiling water is poured into it but the same cup would not crack when immersed in a bath of cold water which is then heated to boiling point;

- (ii) a cat runs with difficulty on a highly polished floor.
- (c) (i) State the pressure law.
- (ii) A bottle is corked when the air in it is at 20°C and 76 cm Hg. The bottle is heated until the pressure is increased by 69 mmHg. Calculate the final temperature of the air in °C.

Question 62

1. A thin converging lens of focal length, f , has an object, O , on and perpendicular to its principal axis. A real image I is produced by the lens as illustrated in the diagram below.
 2. Given that the principal foci of the lens are the points marked f , use the lens formula to show that $f^2 = ab$.
 3. If $a = 16$ cm and $b = 25$ cm, determine f .
 4. Calculate the magnification of the image.
 5. Explain the following terms:
 - Principal focus
 - Optical centre
 - Focal length
-

Question 63

1. List three facts about the acceleration of a free falling object due to gravity.
2. List two objects each which may be considered as projectiles in:
 - Sports
 - Warfare
3. A ball of mass 120g was projected vertically upward from the earth surface with a speed of 25 ms^{-1} . The height of the ball was determined at regular intervals as shown in the table below.

Height h (m)	Time t (s)
11.25	0.5
20.00	1.0
26.25	1.5
30.00	2.0
31.25	2.5

4. Plot a graph with height, h , on the vertical axis and time, t , on the horizontal axis starting both axes from the origin (0,0).

[Draw a smooth curve through the points]

5. Using the graph, determine the height of the ball at $t = 1.7$ s.
6. The combined mass of a bicycle and its rider is 60 kg. Calculate the magnitude of the forward force produced by the bicycle if its speed increases from 10 ms^{-1} to 15 ms^{-1} in 3.0 s.

[Neglect friction and air resistance]

Question 64

Give three observations in support of de-Broglie's assumption that moving particles behave like waves

Question 65

- (a) Define magnetic flux.
- (b) Sketch a diagram to illustrate magnetic lines of force for a bar magnet placed in a uniform earth's magnetic field with its north pole pointing north. Indicate the neutral points.
- (c) State two ways in which the earth's magnetic field is important to life on the earth surface.
- (d) List two applications of electromagnets
- (e) In a certain house, three ceiling fans each of 80 W, an air-conditioner rated 1500 W are operated for 6 hours each day. Seven lamps each rated 40 W are switched on for 10 hours each day. The home theatre 100 W and television set 80 W are also operated each day for 5 hours after which they are kept in the standby mode. In the standby mode, the power consumption is 5% of the power rating. Calculate the:

Total energy consumed in the house for 30 days in kWh;

Cost of operating all the appliances for 30 days at N10.00 per kWh.

Question 66

A ball is projected horizontally from the top of a tall building. State the factors on which the distance at which it strikes the ground from the foot of the building depends.

Question 67

- (a) Define:
 - (i) principal focus for a diverging lens;
 - (ii) principal axis.
- (b) State a situation under which the speed of light can change.

- (c) The separation between the lenses of an astronomical telescope in normal adjustment is 35.0 cm and the angular magnification is 8

Calculate the focal lengths of the lenses.

The lenses separation are adjusted to produce the final image 30 cm from the eye piece lens. Calculate the new separation of the lenses

Sketch a ray diagram to illustrate the image formed in 13(c)(ii) above.

Question 68

Define diffusion.

State two applications of electrical conduction through gases

Question 69

1.(i) What is an eclipse?

(ii) List the three types of eclipse.

2. A student in a lecture theatre can read from the board clearly but requires a pair of spectacles to read from a book.

What eye defect has this student?

What type of lens is needed to correct the eye defect?

The focal length of the lens used to correct this defect is 10 cm. Calculate the power of the lens.

3. A car B moves towards a stationary car A. If B produces an ultrasonic sound at a point and it takes 5.6×10^{-3} s for a beep to be heard in B, calculate the distance between the two cars at that instant. (Speed of sound in air = 340 ms⁻¹)
4. The image of an object is located 9 cm behind a convex mirror. If the magnification produced is 0.6, calculate the focal length of the mirror.

Question 70

1.(i) State two advantages of alcohol over mercury as a thermometric liquid.

(ii) When the bulb of a thermometer is placed in a beaker of hot water, the level of mercury first falls and then rises gradually. Explain this observation.

2. List two factors, other than temperature, that affect the rate of evaporation of a liquid.

3. A block of lead of mass 100kg in a crucible and at a temperature of 40°C was placed in an electric furnace rated 10kW. If the melting point of lead is 320 °C, calculate the:

(i) quantity of heat required to heat the lead to its melting point;

- (ii) additional heat energy required to melt the lead;
- (iii) time taken to supply this additional energy.
- (Specific heat capacity of lead = $120 \text{ JKg}^{-1} \text{ K}^{-1}$)
- (Specific latent heat of fusion of lead = $2.5 \times 10^4 \text{ JK}^{-1}$)
- 4. State two precautions necessary in an experiment to determine the specific latent heat of vaporization of water.

Question 71

- (a) Distinguish between perfectly elastic collision and perfectly inelastic collision.
- (b) Sketch a distance - time graph for a particle moving in a straight line with:
 - (i) uniform speed;
 - (ii) variable speed
- (c) A body starts from rest and travels distances of 120, 300 and 180 m in successive equal time intervals of 12 s. During each interval the body is uniformly accelerated.
 - (i) Calculate the velocity of the body at the end of each successive time interval
 - (ii) Sketch a velocity-time graph for the motion.

Question 72

- (a) State the Heisenberg Uncertainty Principle.
- (b) The product of the uncertainties in Heisenberg Uncertainty Principle is equal to or greater than a constant. State the mathematical expression for this constant.

Question 73

- (a) Write down the names of two particles used in explaining the wave nature of matter
- (b) State the wave characteristics which are exhibited by the particles named in (a) above.

Question 74

1. What is a vector quantity?
2. Three vectors $3 \text{ ms}^{-1} \text{ N } 45^\circ \text{ W}$, $12 \text{ ms}^{-1} \text{ W}$ and $5 \text{ ms}^{-1} \text{ S}$ act at a point.
3. Sketch a vector diagram to illustrate the given information

4. Calculate the resultant of the vectors.
5. In a laboratory experiment to determine the force constant of a spiral spring, the mass on the spring was varied and the corresponding extensions were measured and recorded as shown in the table below.

Mass m (g)	Weight W (N)	Extension e (cm)
50		6.5
100		11.0
150		15.0
200		20.0
250		25.0

1. Copy and complete the table. (Take $g = 10 \text{ ms}^{-2}$)
2. Plot a graph with weight, W , on the vertical axis and extension, e , on the horizontal axis.
3. Using the graph, determine the force constant of the spring.
4. Determine the natural length of the spring if its length was 38.0 cm when loaded with 250 g mass.

Question 75

(a) Explain:

- (i) stopping potential;
- (ii) work function.

(b) State two factors that determine the kinetic energy of ejected electrons in photoelectric

(c) State two uses of a photocell.

(d) When the surface of a certain metal is illuminated with yellow light of wavelength $5.893 \times 10^{-7} \text{ m}$, electrons of stopping potential 0.36 V are liberated. When the surface is illuminated with violet light of wavelength $3.969 \times 10^{-7} \text{ m}$ the stopping potential is 1.38 V.

Calculate the

- (i) Planck's constant, h ;
- (ii) Work function, w ;
- (iii) Threshold frequency f_0 ,

Question 76

(a) With the aid of a labelled diagram, explain.

- (i) node;
- (ii) antinode.

(b) Explain:

- (i) standing wave;
- (ii) persistence of vision.

(c) How does the iris in the human eye respond to:

- (i) Dim light;
- (ii) Excessive brightness?

(d) A wave is represented by the equation $y = 0.20 \sin 0.407t (x - 60t)$ where x distances are measured in centimetre's and time in seconds.

Determine the wave's

- (i) wavelength (ii) frequency.

Question 77

(a) Explain:

- (i) Linear momentum.
- (ii) Impulse of a force

(b) State three instances where Newton's third law of motion is applicable.

(c) A ball of mass 4.0 kg. hits a smooth platform vertically with a speed of 3 ms^{-1} and rebounds with a speed of 2 ms^{-1} . Calculate the impulse experienced by the ball.

(d) A rocket of mass 500 kg carrying 4000 kg of fuel is to be launched vertically. The fuel is consumed at a steady rate of 50 kgs^{-1} , calculate the least velocity of the exhaust

gases if the rocket will just lift off the launching pad immediately after firing
[$g = 10 \text{ ms}^{-1}$]

Question 78

(a) Define:

(i) work;

(ii) power

(b) Distinguish between static friction and dynamic friction

(c) Why do tyres have treads?

(d) A car of mass M starts from rest and moves with a constant acceleration, a . It attains a velocity, v after time t , and maintains this velocity until time t_2 . It is then brought to rest with uniform retardation, d at time t_3 .

(i) Draw and label a velocity-time graph for the motion.

(ii) Deduce an expression for the force on the car during the period of retardation.

Question 79

The force, F , acting on the wings of an aircraft moving through the air of velocity, v , and density, is given by the equation where k is a dimensionless constant and A is the surface area of the wings of the aircraft. Use dimensional analysis to determine the values of k and n .

Question 80

1. Define strain energy.

Write an expression for the energy stored, E , in a stretched wire of original length, l , cross-sectional area, A , extension, e , and Young's modulus, Y , of the material of the wire.

Question 81

(a) Given a retort stand and clamp, a stout pin, a simple pendulum and a pencil, describe how you would use these apparatus to determine the centre of gravity of an irregularly shaped piece of cardboard of a moderate size.

(b) Using a suitable diagram, explain how the following can be obtained from a velocity-time graph:

(i) acceleration;

(ii) total distance covered.

A body at rest is given an initial uniform acceleration of 6.0 ms^{-2} for 20 s after which the acceleration is reduced to 4.0 ms^{-2} for the next 10 s.

The body maintains the speed attained for 30 s.

Draw the velocity-time graph of the motion using the information given above.

From the graph, calculate the:

maximum speed attained during the motion;

total distance traveled during the first 30 s;

average speed during the same time interval as in (ii) above.